EDA Case Study: e-Car Loan Pricing

Workshop 4 — Module 8

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Data Visualization for Data Scientists
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Module Roadmap

1. A new domain: auto loan pricing (208K applications)
2. Apply the same EDA framework to a financial dataset
3. FICO as a continuous predictor discretised into tiers
4. Time-series dimension: spread compression post-2008
5. Structural missingness: MNAR in Previous Rate

Learning Outcome
You will apply the identical EDA workflow from M07 to a completely different domain (consumer finance), demonstrating that the seven-section report structure is domain-agnostic.

The Dataset

e-Car Loan: 208K Applications, 12 Variables

e-Car Loan Dataset: 208K Auto Loan Applications (2002–2012)

Rows	208,087 loan applications
Period	2002–2012 (11 years)
Key numeric	FICO (credit score), Rate (%), Amount ($), Term (months)
Key categorical	Tier (1–5), Car Type (N/U), Outcome (0/1), Partner Bin (1–5)
Derived	Spread = Rate − Cost of Funds (bank margin)
Missing	Previous Rate: 53% missing (new customers = MNAR)
Domain	Auto loan pricing: bank sets Rate based on FICO, Tier, competition

Pro-Tip

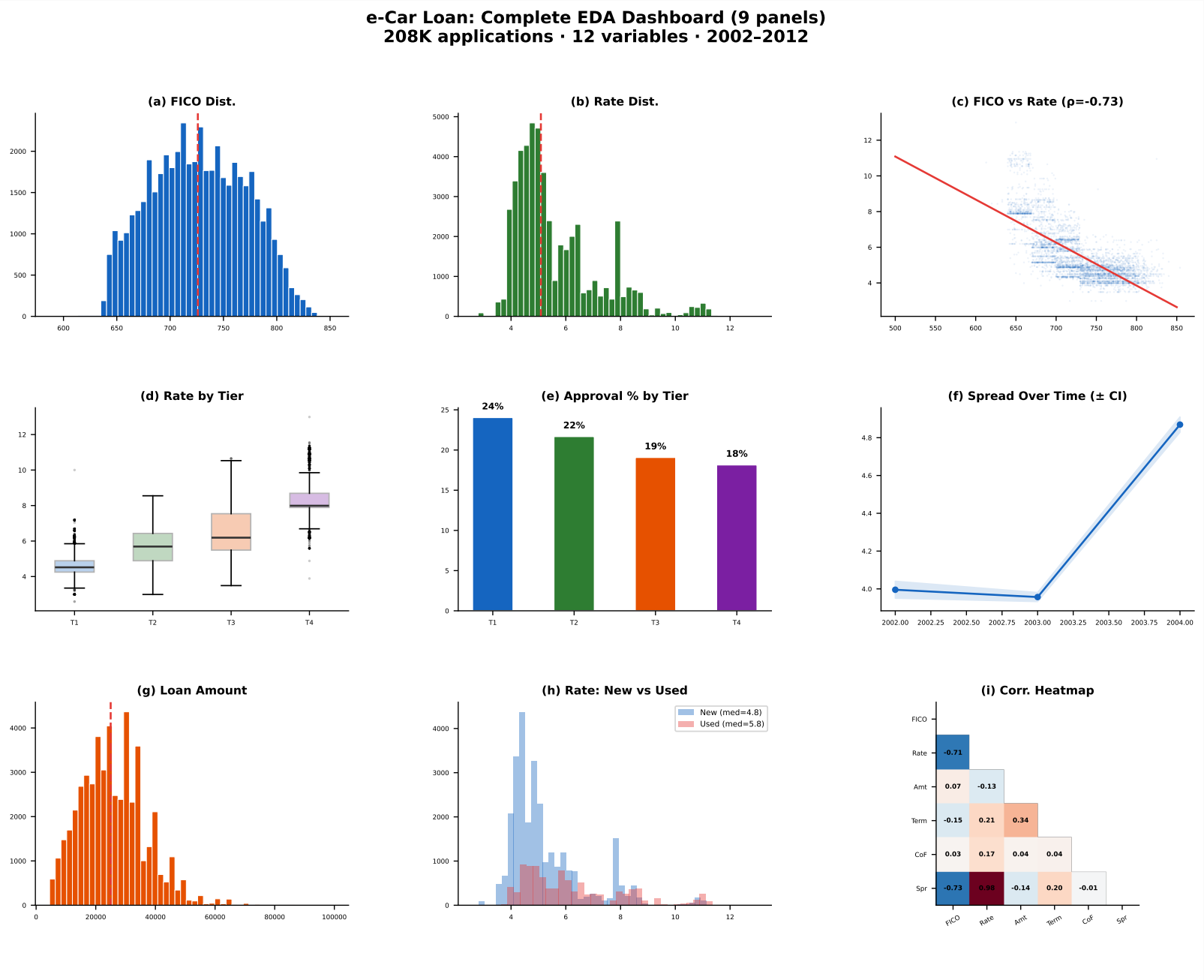
This dataset differs from Google Play Store in three ways: (1) it is **large** (208K vs 10K), requiring sampling for scatter plots; (2) it has a **time dimension** (2002–2012); (3) it contains **structural MNAR** in Previous Rate (new customers simply have no prior rate).

Domain Context: How Auto Loan Pricing Works

Concept	Meaning
FICO	Credit score (300–850). Higher = lower risk.
Tier	Bank's risk bucket (1 = best, 5 = worst). Maps FICO ranges to discrete groups.
Rate	Interest rate offered to borrower (%).
Cost of Funds	Bank's cost of capital (%). Varies with macro rates.
Spread	Rate − Cost of Funds. The bank's **margin**.
Competition Rate	Competitor's rate for same profile.
Outcome	0 = borrower declined offer; 1 = accepted.
Previous Rate	Returning customer's prior rate (missing for new customers).

The Nine-Panel Dashboard

Complete EDA Dashboard

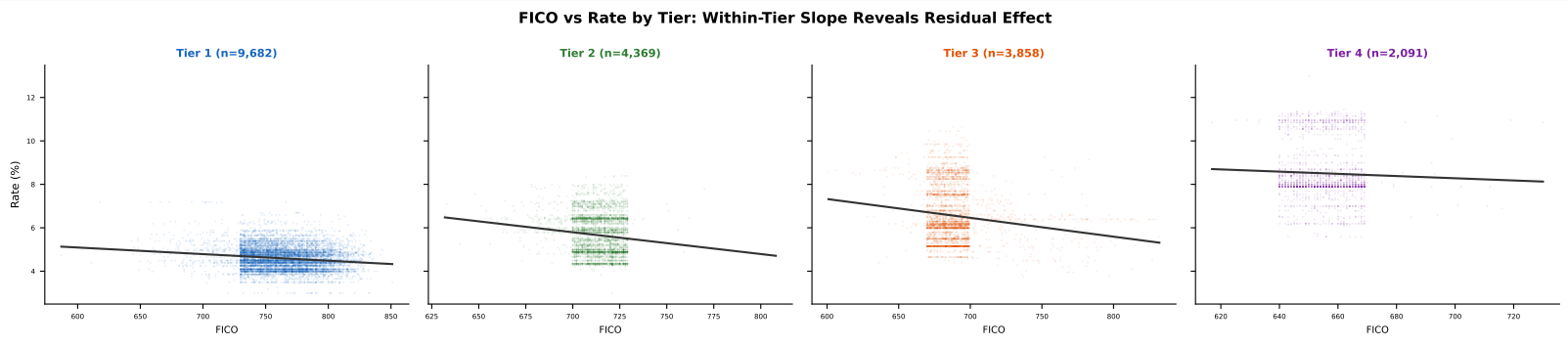


Reading the Dashboard

#	Panel	EDA Question	Insight
(a)	FICO histogram	Shape of credit quality	Bimodal: peaks at $\sim$680 and $\sim$75
(b)	Rate histogram	Shape of pricing	Right-skewed, mode at $\sim$5%
(c)	FICO vs Rate	Key relationship	Strong $\rho \approx -0.65$
(d)	Rate by Tier	Tier = rate driver	Clear step function
(e)	Approval by Tier	Selection effect	Tier 1: 30%, Tier 5: 60%+
(f)	Spread over time	Macro effect	Post-2008 compression
(g)	Amount dist.	Loan size	Right-skewed, median $\sim$\$20K
(h)	New vs Used	Car type effect	Used $\approx$ +1.5pp higher
(i)	Correlation	Full picture	FICO–Rate dominates

Deep-Dives

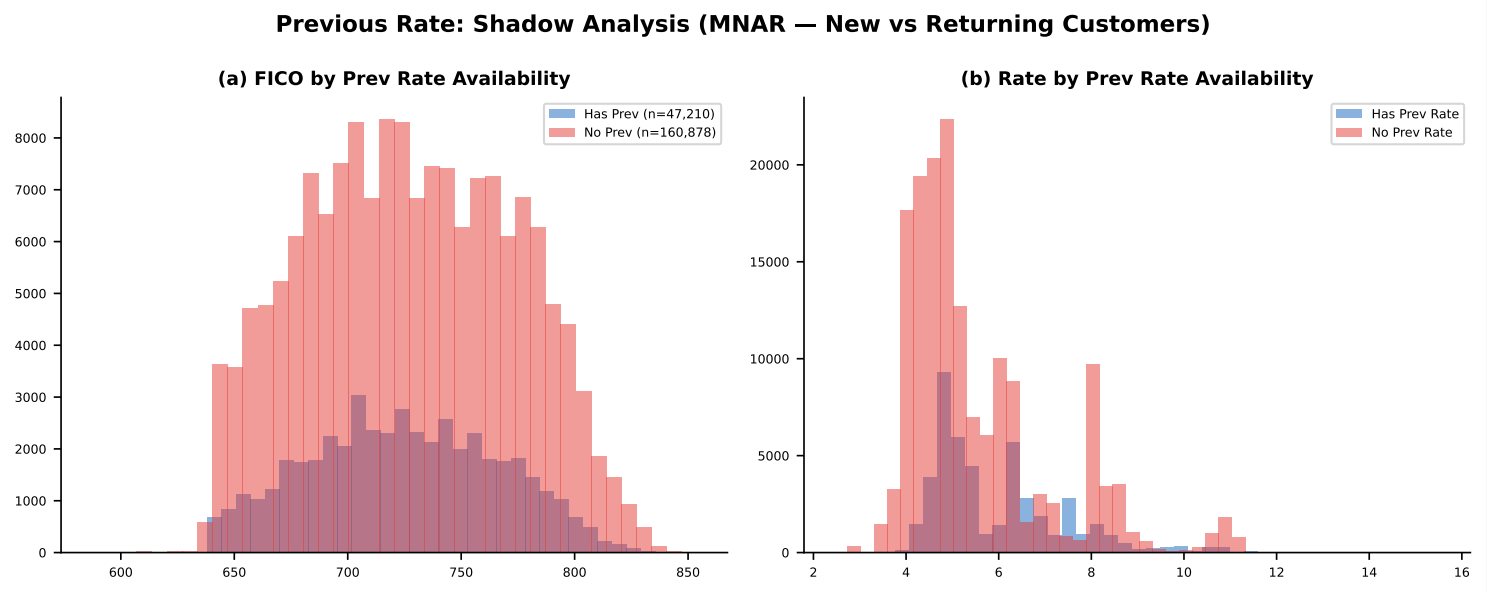
FICO vs Rate: Faceted by Tier



Data Insight

Within each tier, FICO still has a negative slope on Rate. This means the tier system **does not fully capture** the FICO information — there is residual predictive power within tiers. A pricing model could improve by using continuous FICO directly rather than the discretised tier.

Structural Missingness: Previous Rate (MNAR)



Pro-Tip

Previous Rate is missing for 53% of rows — but this is not random. It is structurally absent for **new customers** who have no prior loan history. The shadow analysis shows that the "no prior rate" group has a different FICO distribution (more dispersed) and different offered Rate. This is textbook **MNAR**: the missingness is informative, not ignorable.

Handling Structural MNAR

Approach	When to Use
Create an indicator	Add `is_new_customer = is.na(Previous Rate)`. Treat as a feature, not a gap.
Analyse separately	Split into new vs returning customers. Each group has complete data within its scope.
Don't impute	Mean/median imputation is meaningless: there *is* no previous rate.
Document	Report the 53% structural NA in your EDA report.

Key

insight: structural MNAR is different from accidental missingness. The absence itself carries information ("this is a new customer"), so the correct action is to **encode it as a variable**, not impute it away.

Key Findings

Four Key Findings

e-Car EDA: Four Key Findings

#	Finding	Evidence	Implication
1	**FICO is the dominant predictor of Rate**	Strong negative $\rho = -0.65$ Higher FICO → lower rate	**Tier system discretises a continuous relationship**
2	**Used cars get higher rates than New**	Median rate: Used ≈ +1.5pp above New	**Risk premium for depreciation + default**
3	**Spread compressed post-2008 crisis**	2002: ~4pp → 2010: ~2pp then partial recovery	**Macro environment drives margin more than FICO**
4	**53% of Previous Rate is missing (MNAR)**	New customers have no previous rate → structural NA	**Not imputable; analyse separately**

The Pipeline (R sketch)

```r
ecar <- read_csv("ecar.csv") |>
  rename(CarType = `Car  Type`) |>
  mutate(
    PreviousRate = as.numeric(`Previous Rate`),
    ApproveDate  = mdy(`Approve Date`),
    Year = year(ApproveDate),
    Spread = Rate - `Cost of Funds`,
    is_new = is.na(PreviousRate)  # structural MNAR -> indicator
  ) |>
  filter(FICO >= 300, FICO <= 850, Rate > 0) |>
  mutate(Tier = factor(Tier, levels = 1:5, ordered = TRUE))

# Dashboard: same patchwork structure as M07
(p_fico | p_rate | p_scatter) /
  (p_tier_box | p_approval | p_spread) /
  (p_amount | p_cartype | p_corr) +
  plot_annotation(title = "e-Car Loan: EDA Dashboard",
    tag_levels = "a")
```

Comparing the Two Case Studies

Google Play Store vs e-Car: Same Framework, Different Insights

Dimension	Google Play	e-Car
n	9,360	208,087
Domain	App marketplace	Consumer finance
Key relationship	Reviews $\leftrightarrow$ Rating (weak)	FICO $\leftrightarrow$ Rate (strong)
Missingness	Rating: MCAR (8%)	Previous Rate: MNAR (53%)
Outlier type	Rating floor (real)	Rate spikes (possible errors)
Time dimension	None	11 years (macro cycles)
Imbalance	Free 93% / Paid 7%	Tier 1 (35%) dominates
EDA structure	**Identical 7-section report**	

The same dashboard template, the same cleaning audit, the same finding→evidence→implication format. The **framework is domain-agnostic**; only the content changes.

Synthesis

Key Takeaways

1. The **EDA framework is portable**: same 7-section structure works for app stores and auto loans.
2. **208K rows** requires sampling for scatter plots but enables precise CI estimation.
3. **Structural MNAR** (Previous Rate) should be encoded as an indicator variable, never imputed.
4. The **tier system discretises** a continuous FICO–Rate relationship; within-tier slopes prove residual information.
5. **Macro effects** (2008 crisis) compress spreads — time-series EDA reveals patterns invisible in cross-sections.
6. **Domain context** is essential: "Spread" is meaningless without understanding Cost of Funds.

Essential Readings

- Thomas, L. C. (2009). *Consumer Credit Models: Pricing, Profit, and Portfolios*. Oxford University Press.
- FICO Score documentation: `https://www.myfico.com/credit-education`
- Peng, R. (2016). *Exploratory Data Analysis with R*.

Next Module: EDA Automation & Reproducible Reports (W04-M09)